When Accuracy Follows Position: Diagnosing Training-Induced Shortcuts in LLM Judges

Anonymous ACL submission

Abstract

Pairwise LLM judges increasingly provide both evaluation scores and reward signals, yet their training behavior is often assessed from accuracy under a single response order. We show that this observable can conceal a learned position shortcut introduced by preference-data preprocessing: if a chosen response is always rendered as Response A, response position perfectly predicts the gold verdict. We apply response-order swap as a controlled behavioral intervention that preserves the compared content while changing its slot identity. On Qwen2.5-7B, full GRPO raises original-order accuracy from 80.2% to 94.7%, while mean order-averaged accuracy falls from 77.7% to 74.7% and position-swap consistency falls from 81.5% to 58.7%. Balancing preferred responses across positions instead yields 82.6% order-averaged accuracy and 84.0% position-swap consistency; balanced SFT reaches 90.2% order-averaged accuracy. The same qualitative failure appears across training objectives, proxy rewards, label schemes, learning rates, and model families, although its severity is model-dependent. These results identify a training-induced failure that single-order accuracy cannot detect and motivate response-order intervention as a standard behavioral audit for trained judges.

1 Introduction

LLM judges are increasingly trained to evaluate pairwise preferences and to provide reward signals in post-training pipelines (Zheng et al., 2023; Chiang et al., 2024). These judges are usually optimized and reported by accuracy: whether their verdict matches a gold preference label. Accuracy, however, only identifies the selected label; it does not identify the decision rule that produced it. If prompt construction introduces an artifact that predicts the label, optimization can amplify that artifact while appearing to improve judgment. This creates a basic interpretability question for trained judges: when accuracy improves, has the model learned to compare the responses, or has it learned a shortcut in the training interface?

We identify one such shortcut created by deterministic preference-data preprocessing. Preference datasets often store examples as `chosen`/`rejected`. When a conversion script renders `chosen` as Response A and `rejected` as Response B, the preferred response becomes Response A for every training instance. Position therefore becomes a perfect predictor of the training label under that prompt construction. Any objective that rewards label matching can exploit this regularity by learning a positional policy rather than comparing the responses: apparent accuracy rises, while position-swap consistency collapses (Figure 1).

We diagnose this failure with a controlled behavioral intervention: for each pair, we swap the two responses while preserving their content, then require the judge to give the logically equivalent verdict. This counterfactual perturbation separates apparent accuracy from position-invariant judgment and provides a functional test of whether the verdict follows response content or slot identity. Across SFT, DPO, and GRPO; accuracy-only, decisiveness, confidence-proxy, and composite rewards; and prompt, label, learning-rate, and model-family controls, we find the same accuracy-consistency dissociation. Optimization makes judges look better under the original ordering while making them less stable under position swap. In the strongest failure mode, high original-order accuracy is explained almost entirely by an elevated first-position rate rather than content comparison. The controls change the severity but do not remove the shortcut in our experiments, pointing to the training data layout as the structural source.

This points to a data-level intervention: balance the training data so that preferred responses appear equally often in both positions. The contrast is

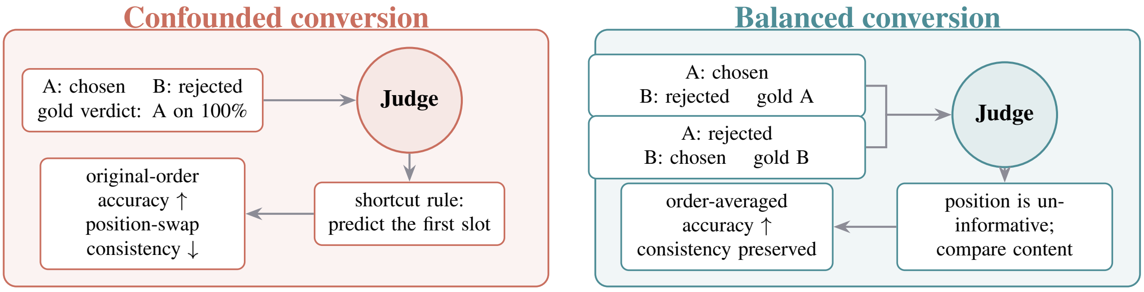

Figure 1: The training-interface shortcut and the data-level intervention. Fixed-order conversion makes position perfectly predictive of the gold verdict, so optimization can raise original-order accuracy while reducing position-swap consistency. Balanced conversion removes this global position-label correlation and forces gains to rely on information beyond a constant slot rule.

clearest in order-averaged accuracy, which gives equal weight to the original and swapped orientations. Under full GRPO, fixed-order training raises original-order accuracy to 94.7% but lowers order-averaged accuracy to 74.7%, below the 77.7% baseline. Balanced GRPO reaches 82.6% order-averaged accuracy, while balanced SFT reaches 90.2%. Thus the apparent gain under the confounded order is not evidence of improved comparison unless it survives response-order averaging.

Our contributions:

- We identify a training-interface shortcut in pairwise LLM judge training: naive `chosen/rejected-to-A/B` preprocessing can make response position a perfect predictor of the gold verdict, allowing trained judges to improve apparent accuracy without learning the intended comparison.
- We apply position swap as a controlled behavioral intervention to trace how this training-time confound changes judge behavior, revealing a repeated gap between single-order and order-averaged accuracy across algorithms, rewards, prompts, labels, learning rates, and model families.
- We trace the failure to the position-label correlation through training controls and a stylized slot-bias analysis, then show that balanced response positions cancel the first-order shortcut gradient and restore both order-averaged accuracy and consistency.

2 Related Work

Bias and Consistency in LLM Judges. Prior work has shown that LLM judges are sensitive to where candidate responses appear in a pairwise prompt (Zheng et al., 2023; Wang et al., 2024; Shi et al., 2024). This position bias interacts with broader evaluator biases, including length preference, self-preference, and reliance on surface features (Wu and Aji, 2023; Koo et al., 2024; Ye et al., 2024; Park et al., 2024; Singhal et al., 2024). Existing work therefore develops metrics and protocols for measuring or mitigating inconsistent judgments, such as swap-based consistency analysis, BT-σ (Qian et al., 2026), inference-time consistency checks (Wang et al., 2025), permutation-aware RL objectives (Xu et al., 2025; Zheng et al., 2026), and selective abstention for uncertain pairwise judgments (Badshah et al., 2026). Our contribution is not the swap test itself or the observation that position bias exists. We use this established intervention to identify a training-time source: deterministic preference-data conversion can make response position a rewarded feature, and optimization can amplify that feature while single-order accuracy appears to improve.

RL Training for Preference Judges. Preference learning has long used human judgments to train language models (Christiano et al., 2017; Stiennon et al., 2020; Ouyang et al., 2022; Bai et al., 2022), and recent work applies similar optimization recipes to the judges themselves. JudgeLRM (Chen et al., 2025) trains reasoning judges with GRPO (Shao et al., 2024); FairJudge (Yang et al., 2026) combines SFT, DPO (Rafailov et al., 2023), and GRPO with consistency-oriented objectives. These approaches treat reward design or algorithmic consistency constraints as the main levers for improving judge behavior. We instead ask whether the data layout makes a non-semantic feature predictive of reward.

Under a position confound, even a correct label reward can be maximized by selecting the first response, so reported accuracy gains must be checked against position-swap consistency.

Behavioral Analysis and Shortcut Learning. Shortcut learning is a well-documented failure mode in which models exploit spurious data correlations rather than acquiring task-relevant features (Geirhos et al., 2020; McCoy et al., 2019; Tu et al., 2020). When a confound perfectly predicts the label, models may learn the confound. Standard mitigations include data augmentation, adversarial training, and causal intervention. In the RLHF setting, reward hacking (Skalse et al., 2022; Gao et al., 2023; Pan et al., 2024) describes a related failure where models exploit flaws in the reward model, and constrained RL (Moskovitz et al., 2023) proposes multi-objective solutions. Our setting highlights a complementary case: the label-matching reward can be correct with respect to the annotation, yet still reinforce a non-semantic route to that label. We show that the tested proxy rewards do not resolve the data-level position confound, while balancing response positions removes the simplest positional reward path.

3 Behavioral Diagnostic Framework

3.1 Pairwise Judge Training and the Position Confound

We study LLM judges as black-box decision systems trained to evaluate pairwise preferences. Given a query q and two candidate responses, the judge produces a rationale and a verdict $v \in \{A, B, \text{tie}\}$. The intended behavior is comparative judgment: the verdict should follow evidence about which response better satisfies the prompt, not incidental artifacts of the prompt interface.

The artifact we study is response position. The position confound arises when preference data are converted into pairwise judge prompts. In standard `chosen`/`rejected` datasets, the preferred response is stored in the `chosen` field. In our RewardBench-to-judge-prompt conversion (Lambert et al., 2024), `chosen` is rendered as Response A and `rejected` as Response B, so the gold label is "A" for every training instance. Position therefore perfectly predicts reward: selecting Response A always earns the maximum accuracy reward.

This makes apparent accuracy ambiguous. A trained judge can improve by learning comparison-relevant evidence, by learning the positional feature, or by mixing both. Evaluation on the same ordering cannot separate these sources, because the shortcut and the intended task produce the same label. The position confound is therefore not an artificial perturbation introduced only at evaluation time; it is inherited from the prompt-construction rule used to build the judge-training data.

3.2 Training Conditions

We compare two training conditions. In the *unbalanced* condition, the prompt conversion renders `chosen` as Response A for every training pair, so the gold verdict is always "A." We then vary the optimization recipe while keeping this confound fixed: SFT, DPO, GRPO with an accuracy reward, and GRPO with decisiveness, confidence-proxy, and composite reward variants. These variants are stress tests for whether common proxy rewards are sufficient to induce comparative judgment under a fixed position-label correlation.

In the *balanced* condition, preferred responses appear in position A for half of the training instances and position B for the other half. Unless otherwise stated, we create this condition by adding a swapped mirror of each training pair and flipping the gold label. We verify in the appendix that the effect is not due to dataset duplication. Once position A is correct exactly half the time, a global first-position rule cannot explain gains over the untrained model.

We use Qwen2.5-7B-Instruct (Qwen Team, 2025a) as the primary model and replicate key conditions on Qwen3-8B (Qwen Team, 2025b) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). The primary Qwen2.5 GRPO runs use LoRA (Hu et al., 2022) over attention projections for 500 steps. Full construction details, reward definitions, cross-model schedules, hyperparameters, and seeds appear in the appendix.

3.3 Position Swap as a Behavioral Intervention

We evaluate each model on 449 held-out test instances, generating responses with temperature 0.1 and max 512 new tokens. For each instance, we generate judgments on both the original and position-swapped order. The intervention preserves the query and response texts while changing only which response occupies each slot. A content-following judge should therefore flip its A/B verdict, whereas a slot-following judge should repeat it. This directly tests whether the verdict follows response content or position.

We report six metrics:

- **Original-order accuracy**: agreement with gold labels on the original ordering.
- **Swap accuracy**: agreement with the flipped gold label after response swapping.
- **Order-averaged accuracy**: the mean of original-order and swap accuracy, giving the two orientations equal weight.
- **Position-swap consistency**: the fraction of instances where the verdict is logically equivalent under position swap. If the model predicts A on the original order, it should predict B on the swapped order, and vice versa. Ties are self-consistent.
- **First-position rate**: the fraction of original and swapped judgments that select the first-listed response.
- **Position bias**: the percentage-point gap between always-first and always-second inconsistent predictions.

Position-swap consistency and order-averaged accuracy are the critical diagnostics because they distinguish performance that survives response order from apparent accuracy under the confounded orientation. In our original-order test set, the preferred response is always first, so original-order accuracy alone is numerically identical to the rate of selecting that preferred slot. We treat ties as self-consistent because a tie verdict is unchanged by swapping response order. Paired tie-only cases are not the driver of the effect: under a stricter tie-as-inconsistent check, the key trained-model consistency comparisons move by at most about one point and conclusions are unchanged. For multi-seed conditions, tables report mean ± standard deviation across seeds. Order-averaged accuracy is computed from the displayed orientation means.

4 Experiments

4.1 Unbalanced Training Creates an Accuracy-Consistency Split

We first ask whether RL training exploits the prompt artifact created by the standard conversion from preference data to judge prompts, where every training label is "A."

Table 1 shows the central dissociation. Training improves original-order accuracy, but this is the orientation on which the preferred response is always first. When both orientations receive equal weight, full GRPO has a lower mean than the untrained model (74.7% versus 77.7%) and substantial seed variation (±3.7), alongside lower position-swap consistency and a sharp increase in the first-position rate. The methods differ in how completely the shortcut takes over: SFT reaches the degenerate endpoint of 100% original-order accuracy but only 50% order-averaged accuracy and zero consistency, while DPO and GRPO retain more comparison signal but still move toward stronger position dependence.

Method	**Orig ↑**	**Swap ↑**	**Avg ↑**	**Con ↑**	**First**	**Bias ↓**
Baseline (no training)	80.2	75.3	77.7	81.5	49.3	2.4
SFT	100.0	0.0	50.0	0.0	100.0	100.0
DPO	94.2	50.3	72.3	54.3	69.4	40.8
GRPO Acc-only	94.7±0.7	55.7±6.4	75.2±2.9	59.8±7.6	68.1±3.6	37.5±7.9
+ Decisive	93.9±1.1	61.0±5.4	77.4±2.1	66.3±6.5	65.0±3.5	31.3±6.8
+ Conf. proxy	95.4±0.5	53.1±7.4	74.3±3.5	56.1±7.9	69.4±4.0	40.3±7.9
+ Both (Full)	94.7±0.9	54.8±8.3	74.7±3.7	58.7±9.1	68.8±4.3	38.6±9.1
GRPO Full (lr=1e-5)	98.9	37.0	67.9	38.1	80.5	61.5
GRPO Full (lr=1e-6)	83.7	76.2	80.0	83.3	51.0	5.6

Table 1: Unbalanced training (gold = "A"). Orig is accuracy under the confounded orientation; Avg averages original and swapped accuracy. Multi-seed entries report mean ± seed standard deviation, including Avg. Training can raise Orig while Avg and consistency fall.

Reward variants and learning rate changes modulate this tradeoff without removing it in our experiments. The best tested reward-only mitigation improves consistency only partially and remains below the untrained baseline. Similarly, conservative learning rates can mask the shortcut by limiting optimization pressure, while the tested Qwen2.5 sweep shows a sharp transition: at larger learning rates, original-order accuracy increases as swap accuracy and position-swap consistency fall (Figure 2). These interventions change when and how strongly the shortcut appears; they do not remove the underlying confound.

These results point back to the data: under the tested training methods, proxy rewards, and learning rates, the most reliable mitigation is to remove the position-label correlation before optimization.

4.2 Balanced Data Restores Position-Invariant Judgment

We therefore train with balanced data (Table 2), assigning preferred responses equally across both slots.

Balanced training reverses the pattern. Models trained on balanced data keep consistency and position bias near the untrained baseline, rather than trading consistency for apparent original-order accuracy (Figure 3). Full GRPO reaches 82.6% order-averaged accuracy, 7.9 points above its unbalanced counterpart and 4.9 points above the untrained model. Balanced SFT reaches 90.2%, showing that direct supervision becomes effective once the target label can no longer be predicted from a single global slot rule.

Method	Orig ↑	Swap ↑	Avg ↑	Con ↑	First	Bias ↓
Baseline (no training)	80.2	75.3	77.7	81.5	49.3	2.4
SFT Balanced	91.3	89.1	90.2	87.1	51.1	2.2
GRPO Acc-only Bal.	84.0±1.8	80.8±1.4	82.4±1.5	84.4±0.4	50.3±0.5	2.3±1.2
GRPO Decisive Bal.	82.9±0.6	79.9±1.5	81.4±0.8	83.4±1.4	50.1±0.5	2.4±1.5
GRPO Conf.-proxy Bal.	83.4±1.1	79.5±1.0	81.5±0.9	83.0±1.3	50.5±0.5	3.0±0.8
GRPO Full Balanced	83.7±1.1	81.4±1.6	82.6±0.8	84.0±0.8	50.0±1.0	1.5±2.2

Table 2: Balanced training (gold = 50% A, 50% B). Multi-seed entries report mean ± seed standard deviation. Data deconfounding keeps the first-position rate near 50%, restores position-swap consistency, and improves order-averaged accuracy over the baseline.

The same comparison also reinterprets the failure of SFT and high-learning-rate GRPO. SFT is catastrophic under unbalanced labels but becomes the strongest balanced-data learner, suggesting that it was efficiently learning the easiest available signal rather than being intrinsically position-biased. High learning rates show the same pattern: harmful under the confound, but useful once the confound is removed. Position preference supports this interpretation. The baseline model is close to position-neutral (49% first-position rate), unbalanced training raises this rate to 69% under full GRPO and 100% under SFT, and balanced training keeps it near 50%. Domain-level breakdowns are reported in the appendix.

4.3 Robustness Summary

We test whether the shortcut is merely an artifact of training duration, inference procedure, prompt wording, label tokens, or generic spurious-feature learning. The same pattern persists across duration, filtering, prompt, and label controls, while a length-confounded control stays near baseline (79.7% consistency, 50.6% first-position rate), suggesting that the severe collapse is specific to the order confound. Checkpoint evaluation shows a transition: early training improves original-order accuracy with little consistency loss, while later optimization trades consistency for a higher first-position rate (Figure 2a). Evaluation-time swap filtering detects but does not repair the failure, and prompt reminders change consistency only marginally (+2.2 pp on Qwen2.5 and -1.1 pp on Mistral).

Changing the output labels from A/B to 1/2 or Left/Right also fails to remove the shortcut (Appendix Table 6). On the original-order prompts, where the preferred response is first-listed, the model continues to select the first-listed response at nearly the same rate under each label scheme, while position-swap consistency remains depressed. The model emits the requested symbols, such as “[[1]]” or “[[Left]],” rather than reverting to “[[A]].” This control rules out a narrow explanation based only on memorizing the literal A token.

4.4 Cross-Model Replication

The shortcut is not specific to the primary Qwen2.5 model. Figure 4 extends the unbalanced-versus-balanced comparison to Qwen3-8B and Mistral-

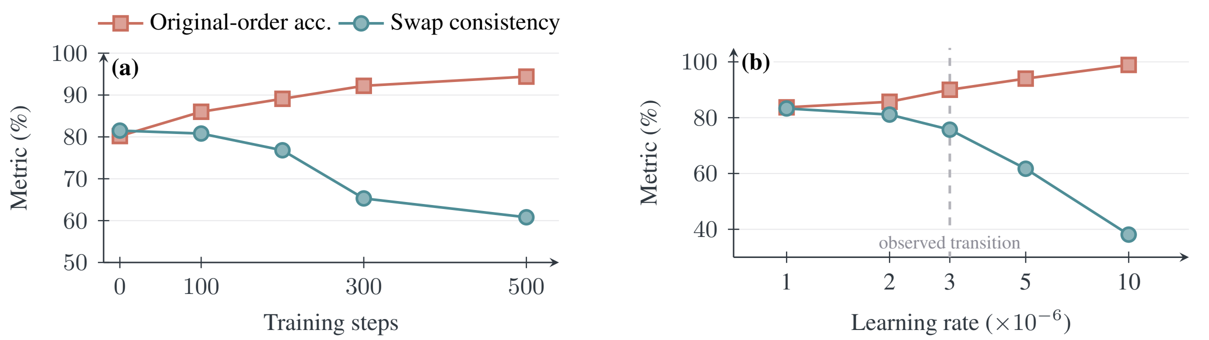

Figure 2: Optimization pressure amplifies the position shortcut in single-seed Qwen2.5 GRPO diagnostics. (a) As training proceeds under the accuracy reward, original-order accuracy rises while position-swap consistency falls. (b) The same tradeoff sharpens across the full-reward learning-rate sweep. The dashed marker denotes an observed transition in the tested grid, not an estimated universal threshold. Detailed checkpoint values and controls are in the appendix.

7B. Severity is model-dependent: Mistral is most affected (98.4% accuracy, 23.0% consistency), Qwen2.5 is intermediate (94.7%/58.7%), and Qwen3 shows only mild GRPO consistency loss (80.4% baseline, 80.0% unbalanced, 82.2% balanced). Qwen3 is less affected in our tested setup, but this three-model comparison does not identify model capability as the cause. Auditing remains necessary: unbalanced Qwen3 training gives a larger apparent original-order gain than balanced GRPO, and unbalanced Qwen3 SFT collapses in the appendix.

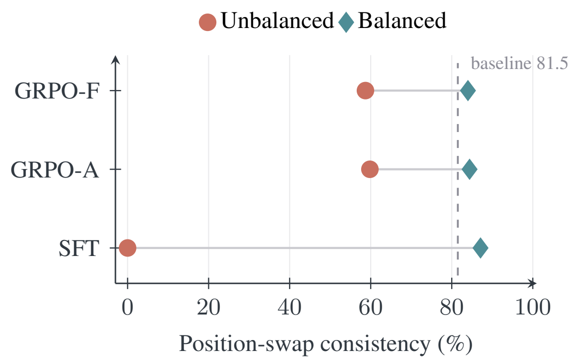

Figure 3: Position-swap consistency under matched unbalanced and balanced data constructions. Coral circles and blue diamonds show the two conditions; connecting segments isolate the data intervention within each training objective. The dashed line is the untrained baseline. GRPO-A and GRPO-F denote accuracy-only and full-composite GRPO.

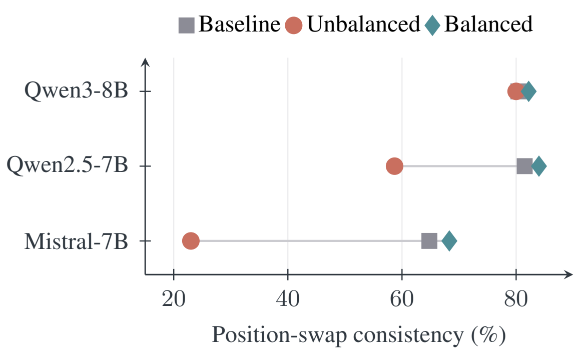

Figure 4: Cross-model GRPO replication. Unbalanced training reduces consistency most strongly on Mistral and Qwen2.5, while balanced data restores or improves consistency across all three tested model families. Exact accuracy values and full controls are in Appendix Table 9.

5 Functional Analysis of the Shortcut

5.1 Why Label Rewards Can Prefer the Shortcut

The experiments above show that the tested reward variants change shortcut severity but do not remove the shortcut. We analyze the mechanism at the functional level: which observable feature can carry predictive signal to the verdict under the training distribution. The data contains two predictive routes to the gold label: comparison evidence E about which response better satisfies the prompt, and response position P (the confound). When P perfectly predicts the gold label, the model has two reward paths: a comparison path ($E \to v^*$) and a positional path ($P \to v^*$). A scalar reward that only checks whether the verdict matches v^* does not identify which path produced the answer.

A stylized slot-bias model illustrates this pressure. It is not a derivation of the GRPO update, but a local analysis of how label imbalance creates a direct optimization direction for a position feature. Let $y \in \{+1, -1\}$ indicate whether the first or second response is preferred, and let $f_\theta(x) = d_\theta(x) + b$, where $d_\theta(x)$ is a comparison score and b is a global first-position bias. Under logistic loss, near $d_\theta(x) = b = 0$,

$$\frac{\partial \mathcal{L}}{\partial b} \approx -\tfrac{1}{2}(2\alpha - 1), \qquad \alpha = \Pr(y = +1).$$

Thus fully unbalanced data ($\alpha = 1$) induces a first-order gradient toward choosing the first response, while balanced data ($\alpha = 0.5$) cancels this shortcut gradient. Appendix Q gives the full derivation.

This also clarifies why the tested proxy rewards fail. Decisiveness penalizes ties but does not require position invariance. Our confidence proxy applies a Brier-shaped score to a fixed fallback confidence when no confidence is emitted; it therefore remains largely aligned with verdict correctness and does not make position dependence costly. The composite reward remains satisfiable by the shortcut. Balanced data changes the training distribution instead: it removes the global $P \to v^*$ correlation, so position no longer reliably predicts reward.

The confound-ratio diagnostic in Appendix R provides a boundary check. Intermediate single-seed ratios are noisy and non-monotonic, but the fully unbalanced endpoint shows a much larger consistency loss than the balanced reference.

5.2 Single-Order and Order-Averaged Accuracy

Original-order accuracy and order-averaged accuracy answer different questions in this setting. The former evaluates only the orientation used by the confounded conversion, where the preferred response is always first. The latter averages the two logically equivalent orientations and therefore prevents an always-first policy from scoring above chance. This is a descriptive contrast, not a causal decomposition of model knowledge.

For Qwen2.5, the distinction reverses the apparent conclusion. Full GRPO raises original-order accuracy from 80.2% to 94.7%, yet its order-averaged accuracy is 74.7%, below the 77.7% untrained baseline. After data balancing, full GRPO reaches 82.6% and SFT reaches 90.2% order-averaged accuracy. The result isolates the paper's practical claim: an accuracy gain is credible only when it persists across both response orders.

5.3 A Qualitative Black-Box Trace

One inconsistent prediction exposes the same failure at the level of generated rationales (Table 3). After the two physical responses are exchanged, the judge correctly describes the newly assigned slot contents but still selects Response A. The example does not establish frequency; it shows why a fluent, content-aware rationale is not by itself evidence that the verdict is position-invariant.

Order	Rationale excerpt	Verdict
Original	"Assistant A's response is more directly relevant ... Assistant B's answer discusses environmental protection laws broadly."	A
Swapped	"Assistant A provides a more detailed and structured response ... broader environmental protection context. Assistant B's response ... directly addresses animal welfare."	A

Table 3: Illustrative inconsistent prediction from full GRPO (example 2790). The same physical answers are swapped, but the verdict remains A.

5.4 External Check on Published RL-Trained Judges

As an external check, we evaluate JudgeLRM's publicly released checkpoints (Chen et al., 2025) with the same position-swap diagnostic (Table 4). JudgeLRM-7B drops from 81.3% original-order accuracy to 66.1% swap accuracy, while JudgeLRM-3B drops from 76.8% to 55.9%. Their two-order first-position rates are close to neutral, but positive bias gaps indicate that inconsistent predictions more often favor the first position than the second.

These results do not imply that all gains in prior judge training are shortcuts. They show that standard accuracy can hide a position-dependent component even for independently released RL-trained judges, motivating swap-consistency reporting.

Model	Orig Acc	Swap Acc	Con	First-pos	Bias
JudgeLRM-7B	81.3	66.1	75.7	53.1	13.4
JudgeLRM-3B	76.8	55.9	67.9	56.7	15.4

Table 4: External check on public RL-trained judges. Both checkpoints show an original/swap accuracy gap and positive first-position bias among inconsistent predictions.

6 Discussion

The position shortcut highlights a failure mode that is easy to miss when judge training is evaluated only by accuracy. The reward need not be noisy or misspecified with respect to the annotation: a label-matching reward can reinforce the wrong feature when the prompt construction makes that feature perfectly predictive. In this sense, the failure is related to reward hacking (Skalse et al., 2022; Gao et al., 2023) but not identical to it. The policy is not exploiting a flawed reward model; it is exploiting a confounded route to the correct reward.

Our diagnosis is behavioral rather than circuit-level. Position swap establishes a functional dependence of the verdict on slot identity, while data balancing intervenes on the source of that dependence. Together with the learning-rate and confound-ratio trajectories, these interventions explain when the shortcut becomes reward-predictive and how strongly training exploits it, without claiming a unique internal representation or circuit.

This distinction matters for mitigation. Adding reward terms can help only if those terms make the shortcut costly. In our setting, decisiveness and the fixed-confidence proxy remain compatible with always choosing the rewarded position, so they do not distinguish comparison-based judgments from positional ones. Data balancing removes the global position-label correlation before optimization by making each position correct equally often. Algorithm-level methods such as EIS-GRPO (Xu et al., 2025) and PA-GRPO (Zheng et al., 2026), as well as evaluation-time methods such as SCOPE (Badshah et al., 2026), are complementary: they can enforce or test permutation awareness, while data deconfounding removes the simplest positional reward path at the source.

The practical implication is straightforward. Pairwise judge training should randomize or balance response order before any supervised, preference, or RL objective is applied, and reported accuracy should be paired with position-swap consistency. This recommendation is not specific to our RewardBench conversion. Any pipeline that converts `chosen/rejected` data into fixed-order judge prompts can introduce the same confound unless it explicitly randomizes position. Since LLM judges are increasingly used as both evaluators and reward signals, failing to audit this confound can propagate ordering bias into downstream model selection and post-training loops.

6.1 A Minimal Audit Protocol

A minimal audit has four steps: (1) measure position–label association after data conversion; (2) evaluate paired held-out examples in both orientations under identical decoding; (3) report the five diagnostics for untrained and trained checkpoints; and (4) rerun the same recipe with balanced positions. Swapping reveals positional dependence, while balancing tests whether the training-time correlation induced it.

Interpret the metrics jointly. Robust gains improve both orientations without sacrificing consistency. Original-only gains with falling swap accuracy and consistency, plus a rising first-position rate, indicate position dependence; high consistency with low order-averaged accuracy can still be wrong. Use order-averaged accuracy for selection. Because balancing cannot exclude other shortcuts, repeat the paired audit after mitigation.

Limitations

Our experiments use 7–9B parameter open-source models from three families (Qwen2.5, Qwen3, Mistral) with LoRA training. We have not tested frontier closed-source models or full fine-tuning. We validate primarily on RewardBench converted into pairwise judge prompts, so empirical validation on additional training and evaluation suites would strengthen external validity. The position confound can arise in any pipeline that maps a `chosen` field to a fixed prompt position, but the severity of shortcut learning may depend on the dataset, prompt template, model, and optimization recipe. Our analysis identifies functional reliance on response position but does not localize the internal representation or circuit that implements it. We do not compare against permutation-aware RL methods (EIS-GRPO, PA-GRPO) which address position bias at the algorithm level rather than the data level. Position-swap consistency measures positional robustness but does not directly measure alignment with human judgment quality. Each orientation is generated once at temperature 0.1, so measured consistency also contains residual sampling variability; the large collapses are unlikely to be explained by this alone, but small cross-model differences should be interpreted cautiously.

7 Conclusion

We showed that accuracy is not, by itself, a certificate of judgment ability for trained LLM judges. When preference data are converted into fixed-order pairwise prompts, response position can become a perfectly reward-predictive artifact. Standard optimization can then improve apparent accuracy by exploiting the training interface rather than comparing the responses. Response-order intervention makes this otherwise hidden decision rule observable. In our experiments, the position shortcut appears across SFT, DPO, and GRPO, persists under the tested reward variants, prompt instructions, label changes, and learning-rate controls, with substantial variation across three model families.

The remedy is simple but diagnostic: remove the position-label correlation before optimization. Balanced response positions cancel the shortcut gradient, restore position-swap consistency, and retain non-positional accuracy gains in our setting. More broadly, our results suggest that RL-trained judge evaluations should report not only accuracy but also whether the verdict is invariant to swapping response order. Because this audit only requires evaluating the same examples after a response-order swap, it is cheap relative to training and practical to report by default. We therefore recommend response-order invariance, alongside order-averaged accuracy, as a release criterion for pairwise judge checkpoints and as a routine diagnostic throughout data conversion, model training, and checkpoint selection. At minimum, a release audit should verify training-prompt position–label balance, compare paired original and swapped judgments under identical decoding, and repeat the check after checkpoint selection. These tests separate stable comparison gains from presentation-order artifacts. For pairwise judges, accuracy is evidence only when it survives swapping.

References

Sher Badshah, Ali Emami, and Hassan Sajjad. 2026. SCOPE: Selective conformal optimized pairwise LLM judging. *Preprint*, arXiv:2602.13110.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, and 12 others. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *Preprint*, arXiv:2204.05862.

Nuo Chen, Zhiyuan Hu, Qingyun Zou, Jiaying Wu, Qian Wang, Bryan Hooi, and Bingsheng He. 2025. JudgeLRM: Large reasoning models as a judge. *Preprint*, arXiv:2504.00050.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. 2024. Chatbot arena: An open platform for evaluating LLMs by human preference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 8359–8388. PMLR.

Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning*.

Robert Geirhos, J"orn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7b. *Preprint*, arXiv:2310.06825.

Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, and Dongyeop Kang. 2024. Benchmarking cognitive biases in large language models as evaluators. *Preprint*, arXiv:2309.17012. Published at ACL 2024.

Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2024. RewardBench: Evaluating reward models for language modeling. *Preprint*, arXiv:2403.13787.

R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

Ted Moskovitz, Aaditya K. Singh, DJ Strouse, Tuomas Sandholm, Ruslan Salakhutdinov, Anca D. Dragan, and Stephen McAleer. 2023. Confronting reward model overoptimization with constrained RLHF. *Preprint*, arXiv:2310.04373.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*.

Jane Pan, Tianyu Shih, Cameron Goldie, and Robert Geirhos. 2024. Spontaneous reward hacking in iterative self-refinement. *Preprint*, arXiv:2407.04549.

Ryan Park, Rafael Rafailov, Stefano Ermon, and Chelsea Finn. 2024. Disentangling length from quality in direct preference optimization. *Preprint*, arXiv:2403.19159.

Mengjie Qian, Guangzhi Sun, Mark J. F. Gales, and Kate M. Knill. 2026. Who can we trust? LLM-as-a-jury for comparative assessment. *Preprint*, arXiv:2602.16610.

Qwen Team. 2025a. Qwen2.5 technical report. *Preprint*, arXiv:2412.15115.

Qwen Team. 2025b. Qwen3 technical report. *Preprint*, arXiv:2505.09388.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *Preprint*, arXiv:2402.03300.

Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, and Soroush Vosoughi. 2024. Judging the judges: A systematic study of position bias in LLM-as-a-judge. *Preprint*, arXiv:2406.07791. AACL-IJCNLP 2025.

Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2024. A long way to go: Investigating length correlations in RLHF. In *Conference on Language Modeling*.

Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems*.

Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. 2020. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems*.

Lifu Tu, Garima Lalwani, Spandana Gella, and He He. 2020. An empirical study on robustness to spurious correlations using pre-trained language models. *Transactions of the Association for Computational Linguistics*, 8:621–633.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Yidong Wang, Yunze Song, Tingyuan Zhu, Xuanwang Zhang, Zhuohao Yu, Hao Chen, Chiyu Song, Qiufeng Wang, Cunxiang Wang, Zhen Wu, Xinyu Dai, Yue Zhang, Wei Ye, and Shikun Zhang. 2025. TrustJudge: Inconsistencies of LLM-as-a-judge and how to alleviate them. *Preprint*, arXiv:2509.21117.

Minghao Wu and Alham Fikri Aji. 2023. Style over substance: Evaluation biases for large language models. *Preprint*, arXiv:2307.03025.

Austin Xu, Yilun Zhou, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. 2025. J4R: Learning to judge with equivalent initial state group relative policy optimization. *Preprint*, arXiv:2505.13346.

Bo Yang, Lanfei Feng, Yunkui Chen, Yu Zhang, Xiao Xu, and Shijian Li. 2026. FairJudge: An adaptive, debiased, and consistent LLM-as-a-judge. *Preprint*, arXiv:2602.06625.

Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Chawla, and Xiangliang Zhang. 2024. Justice or prejudice? quantifying biases in LLM-as-a-judge. *Preprint*, arXiv:2410.02736.

Jinquan Zheng, Jia Yuan, Jiacheng Yao, Chenyang Gu, Pujun Zheng, and Guoxiu He. 2026. Mitigating selection bias in large language models via permutation-aware GRPO. *Preprint*, arXiv:2603.21016. Accepted to ACL 2026 Main Conference.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and chatbot arena. *Preprint*, arXiv:2306.05685.

A Training Details

Table 5 lists the shared training hyperparameters used unless otherwise stated.

Reward definitions. The accuracy reward is binary label match: $R_{\text{acc}} = \mathbb{1}[v = v^*]$. The decisiveness proxy is $R_{\text{dec}} = 0.5\mathbb{1}[v \neq \text{C}]$. The confidence proxy is a Brier-shaped score $R_{\text{conf}} = 1 - (\tilde{c} - \mathbb{1}[v = v^*])^2$, where the parser uses an emitted confidence if present and otherwise sets $\tilde{c} = 0.8$ for A/B or 0.5 for C. The composite reward is $R = \alpha R_{\text{acc}} + \beta R_{\text{dec}} + \gamma R_{\text{conf}}$, with $\alpha = 1.0$, $\beta = 0.5$, and $\gamma = 0.3$. Because the prompt requests only a verdict and none of the 119 saved evaluation summaries records an explicit confidence, we treat this component as a fixed-confidence proxy, not a calibration measure.

Hyperparameter	Value
Base model	Qwen2.5-7B-Instruct
LoRA rank / α	64 / 128
Target modules	q, k, v, o projections
GRPO group size	8
Batch size	4 (grad accum 4)
Learning rate	5×10^{-6} (default)
LR scheduler	Cosine decay
Max prompt / response	2048 / 1024 tokens
Training steps	500 (primary Qwen2.5)
Reward weights	1.0, 0.5, 0.3
Seeds	42, 123, 456, 789
Hardware	1× H200 (143GB) per run
Training time	∼2.5h per run

Table 5: Primary Qwen2.5 training hyperparameters. For the learning-rate study, we additionally test 1×10^{-5}, 3×10^{-6}, 2×10^{-6}, and 1×10^{-6}. Cross-model controls use the run lengths recorded with their released experiment configurations.

B Data Construction

RewardBench (Lambert et al., 2024) provides 2,985 preference pairs drawn from multiple evaluation subsets. With seed 42, we randomly shuffle instance indices and partition them into 2,089 training, 447 reserved validation, and 449 test instances. The validation slice was not saved or used for early stopping, hyperparameter selection, or model selection in the reported runs. For analysis only, we aggregate test subsets into five slices—Chat, Reasoning, Safety, Code, and Adversarial—as detailed in Appendix P. In our unbalanced judge-prompt conversion, the `chosen` field is rendered as Response A for every training and test instance.

For unbalanced training, we use this data as-is. For balanced training, we include both orientations of each training pair: the original instance with gold label A and a swapped mirror with gold label B. This yields 4,178 balanced training prompts and ensures that position A is preferred in exactly half the training instances. We also run a non-duplicated balanced control with 2,089 prompts in Appendix R.

Position bias in preference-data conversion. The "always A" artifact is a property of the conversion rule, not an unavoidable property of RewardBench or the `chosen`/`rejected` schema. Any converter that places `chosen` in a fixed judge-prompt slot creates the confound; randomizing or balancing the slots prevents it. We therefore make no claim that existing pipelines using other preference datasets necessarily contain the same artifact.

For SFT, we pair each judge prompt with a fixed generic rationale template naming the gold slot, followed by "[[A]]" or "[[B]]." For DPO, the chosen and rejected outputs use the same generic slot-specific templates with the correct and opposite verdicts, respectively. These targets test whether the optimization objective learns the label-position regularity; they are not intended as high-quality rationale supervision.

SFT 100% accuracy sanity check. The SFT result (100% original-order accuracy, 0% consistency on unbalanced data) is extreme but expected. Every target contains both a generic rationale favoring Assistant A and the verdict "[[A]]," so either part of the target exposes the same slot regularity. Balanced SFT removes the global constant-label solution by using A and B equally often. Its improvement therefore shows that SFT can learn a rule beyond always selecting one slot; it should not be read as evidence about rationale quality.

For evaluation, we always test both the original and position-swapped version of each test instance, using the swap rate as our consistency metric.

Artifact scope. The anonymous artifact includes the exact conversion, SFT, DPO, GRPO, evaluation, and aggregation scripts; the saved per-example predictions used for every reported table; and a CPU-only command that regenerates the summaries and bootstrap checks. Model checkpoints and licensed source examples are omitted for size and redistribution reasons. Machine-specific paths in stored aggregate metadata are redacted without changing metric values.

C Prompt Template

The following is the training and evaluation template; line breaks are wrapped for display.

```
Please act as an impartial judge and evaluate the quality
of the responses provided by two AI assistants to the user
question displayed below. You should choose the assistant
that follows the user's instructions and answers the user's
question better. Your evaluation should consider factors
such as the helpfulness, relevance, accuracy, depth,
creativity, and level of detail of their responses.

Begin your evaluation by comparing the two responses and
provide a short explanation. Avoid any position biases and
ensure that the order in which the responses were presented
does not influence your decision. Do not allow the length of
the responses to influence your evaluation. Be as objective
as possible.

After providing your explanation, output your final verdict
by strictly following this format: "[[A]]" if assistant A is
better, "[[B]]" if assistant B is better, and "[[C]]" if it
is a tie or you cannot determine a winner.

[User Question]
{question}

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]
```

Output parsing and format handling. Evaluation parses `[[A]]`, `[[B]]`, or `[[C]]`; a parse failure is counted as incorrect and inconsistent. The parser also accepts an optional numeric confidence, but all 119 saved evaluation summaries report zero explicit-confidence outputs. During GRPO training, an unparseable completion is mapped to C with fallback confidence 0.5; since all training labels are A/B, it receives zero accuracy and decisiveness reward but a confidence-proxy value of 0.75. Ties are treated as self-consistent when both orientations produce C. A stricter tie-as-inconsistent check changes the key trained-model consistency comparisons by at most about one point and does not alter the conclusions.

D Prompt and Label Controls

Prompt-level mitigation ("evaluate based on content quality only; ordering is random") only marginally changes consistency: Qwen2.5 moves from 61.7% to 63.9%, while Mistral moves from 20.0% to 18.9%. We also evaluate unbalanced-GRPO models with alternative label schemes: "Response 1/2" and "Response Left/Right." The shortcut persists across all prompt and label variants and all three model families we test (Table 6). Zero parse failures confirm that the Qwen2.5 and Mistral models follow the relabeled format rather than reverting to "[[A]]." For Qwen3, the reported prompt/label-control runs use thinking disabled and retain parse failures below 1.3%; without this setting, parse failures rise to 39–42% and the control is not interpretable.

Model	Control	Acc	Con	Orig.	Parse
Qwen2.5	A/B	94.0	61.7	94.0	0.0
	Anti	93.8	63.9	93.8	0.0
	1 / 2	94.7	60.8	94.7	0.0
	L/R	94.2	59.9	94.2	0.0
Qwen3	A/B	88.4	78.8	88.4	1.1
	Anti	88.0	80.2	88.0	0.7
	1 / 2	88.4	81.7	88.4	1.2
	L/R	85.7	83.7	85.7	1.0
Mistral	A/B	98.9	20.0	98.9	0.0
	Anti	98.7	18.9	98.7	0.0
	1 / 2	99.6	12.0	99.6	0.0
	L/R	99.1	15.6	99.1	0.0

Table 6: Prompt and label controls. Anti is the ordering-randomized anti-shortcut prompt; L/R denotes Left/Right labels. Orig. is original-order first-selection; Parse is the two-order parse-failure rate.

E Evaluation-Time Swap Filtering

Post-hoc swap filtering can identify examples on which the original and swapped predictions disagree, but it does not repair the underlying model on those examples. We keep only pairs whose predictions are logically consistent after swapping positions. This produces a high-accuracy retained subset, but coverage collapses exactly when the shortcut is severe: unbalanced SFT retains 0.0% of examples, and unbalanced full GRPO retains only 58.7% (Table 7). A random fallback on discarded examples reduces unbalanced full GRPO to 74.9% expected accuracy, while balanced full GRPO keeps 84.0% coverage and 82.6% expected fallback accuracy. Thus swap filtering is useful as a diagnostic or selective abstention rule, but data balancing addresses the source of the shortcut.

Setting	Acc	Cov.	Ret.	Rand.
Baseline	80.2	81.5	84.2	77.8
SFT unbal	100.0	0.0	–	50.0
SFT bal	91.3	87.1	96.2	90.2
GRPO-A unbal	94.7	59.8	92.4	75.3
GRPO-F unbal	94.7	58.7	92.3	74.9
GRPO-F bal	83.7	84.0	88.8	82.6

Table 7: Post-hoc swap filtering on Qwen2.5. Cov. is retained coverage; Ret. is accuracy on retained examples; Rand. assumes a random binary fallback on discarded examples. GRPO rows report seed means.

F Length-Confound Control

To separate position shortcuts from generic shortcut learning, we include a non-positional confound control in which the training preference is correlated with response length rather than response order. This control does not produce the same failure mode: length-confounded GRPO remains close to the untrained baseline in both consistency and first-position rate (Table 8). In contrast, position-confounded GRPO-A increases original-order accuracy to 94.7% while reducing consistency to 59.8% and pushing the first-position rate to 68.1%. This suggests that the severe collapse we study is tied to the globally predictive position feature introduced by preference-data conversion, not simply to the presence of any spurious training feature.

Setting	Acc	Swap	Con	First
Baseline	80.2	75.3	81.5	49.3
Position-conf. GRPO-A	94.7	55.7	59.8	68.1
Length-conf. GRPO	80.9	75.8	79.7	50.6

Table 8: Length-confound control on Qwen2.5. Swap is swapped-order accuracy; First is the two-order first-position rate. The two GRPO rows report three-seed means.

G Cross-Model Validation

To verify that our findings generalize beyond a single model family, we replicate experiments on Qwen3-8B (Qwen Team, 2025b) (a newer-generation Qwen model) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) (a completely different architecture and pre-training corpus from Mistral AI). Table 9 reports the full cross-model comparison. For Qwen3, we use the disable-thinking baseline run so that parsing behavior matches the rest of the Qwen3 evaluations.

Method	Data	Acc	Con
Qwen2.5-7B-Instruct (primary)			
Baseline	—	80.2	81.5
SFT	unbal	100.0	0.0
DPO	unbal	94.2	54.3
GRPO	unbal	94.7±0.9	58.7±9.1
SFT	bal	91.3	87.1
GRPO	bal	83.7±1.1	84.0±0.8
Qwen3-8B (cross-generation)			
Baseline	—	86.0	80.4
SFT	unbal	100.0±0.0	0.0±0.0
DPO	unbal	89.3	81.3
GRPO	unbal	88.7±0.3	80.0±1.0
SFT	bal	88.5±1.1	89.1±0.3
DPO	bal	88.6	85.3
GRPO	bal	87.4±0.8	82.2±0.6
Mistral-7B-Instruct-v0.3 (cross-family)			
Baseline	—	65.7	64.8
SFT	unbal	100.0	0.0
SFT	bal	92.2	84.9
GRPO	unbal	98.4±0.4	23.0±3.6
GRPO	bal	81.1±0.9	68.3±3.6

Table 9: Cross-model validation across three model families. The position shortcut replicates, but severity is model-dependent: Mistral shows the most extreme collapse, Qwen2.5 is intermediate, and Qwen3 shows the mildest GRPO consistency loss under disable-thinking evaluation.

Shortcut severity is model-dependent. Mistral-7B has the lowest baseline original-order accuracy (65.7%) and the largest trained shortcut (98.4% accuracy, 23.0% consistency). Qwen2.5-7B (80.2% baseline) is intermediate (94.7%/58.7%), whereas Qwen3-8B has the highest original-order baseline (86.0%) and much milder GRPO consistency damage (80.4% baseline → 80.0% unbalanced); balanced GRPO reaches 82.2%. These three observations are consistent with several explanations, including different initial comparison behavior and model-specific optimization response, but they do not identify capability as the cause. The Qwen3 SFT collapse to 0.0% consistency also shows that mild GRPO sensitivity is not immunity. Balanced data is the more direct intervention because it removes the position-label correlation instead of relying on a particular model–optimizer interaction.

The models also differ in the *speed* of shortcut emergence. Checkpoint-level evaluation shows that Mistral-7B is already strongly position-dependent by step 100 (98.7% accuracy, 20.3% consistency), while Qwen2.5-7B changes more gradually through 500 steps (84.6%/81.7% at step 100 → 94.0%/61.7% at step 500). Because training schedules and models differ, this comparison is descriptive rather than a controlled estimate of capability effects. Practically, it motivates auditing early checkpoints rather than waiting until the end of training.

A flatter learning-rate sensitivity pattern also appears on Qwen3-8B. Consistency stays in the high-70s to low-80s across its accuracy-reward sweep (Table 10), compared with the sharp Qwen2.5-7B decline under the full reward. Because both the model and reward differ, the curves establish configuration-specific sensitivity rather than a pure model-capability effect.

LR	Qwen2.5-7B		Qwen3-8B	
	Acc	Con	Acc	Con
1e-6	83.7	83.3	89.1	82.4
2e-6	85.7	81.1	87.8	81.5
3e-6	90.0	75.7	87.8	82.6
5e-6	94.0	61.7	88.4	78.8
1e-5	98.9	38.1	90.2	78.4

Table 10: Cross-model LR sensitivity. The Qwen2.5 sweep uses the full reward, while the Qwen3 sweep uses the accuracy reward; rows are single-seed controls. Qwen2.5 loses about 45 pp of consistency across the range, compared with about 4 pp for Qwen3, but this is not a controlled model-only comparison.

Multi-objective rewards also fail to reliably restore consistency on Qwen3-8B: the decisiveness proxy raises consistency from 78.8% to 82.0%, while the confidence proxy lowers it to 78.0%. The composite reward remains below balanced GRPO under the tested weights. These results show that the tested proxy rewards are insufficient for this data-level confound across tested configurations.

All experiments use NVIDIA H200 GPUs (143GB). Total compute: approximately 200 H200-GPU-hours across all experiments.

H External Check on Published RL-Trained Judges

As an external check, we evaluate JudgeLRM's publicly released checkpoints (Chen et al., 2025) with the same position-swap diagnostic (Table 11). JudgeLRM-7B achieves 81.3% original-order accuracy but 66.1% swap accuracy and 75.7% consistency, while JudgeLRM-3B achieves 76.8% original-order accuracy, 55.9% swap accuracy, and 67.9% consistency. These results do not imply that all gains in prior judge training are shortcuts; they show that standard accuracy can hide a position-dependent component.

Model	Orig Acc	Swap Acc	Con	First-pos	Bias
JudgeLRM-7B	81.3	66.1	75.7	53.1	13.4
JudgeLRM-3B	76.8	55.9	67.9	56.7	15.4

Table 11: Published RL-trained judges also show a position-dependent component under the swap diagnostic. Bias is the always-first minus always-second inconsistent rate.

I Multi-Seed Stability

The primary GRPO pattern is stable across random seeds rather than driven by a single run. Table 12 reports individual seed results for representative unbalanced and balanced conditions. Unbalanced runs maintain high original-order accuracy but low consistency and an elevated first-position rate; balanced runs keep consistency high and the first-position rate near neutral.

Method	Seed	Acc	Con	First-pos
Acc-only unbal	s1	94.4	60.8	67.4
	s2	95.6	51.7	72.0
	s3	94.2	66.8	64.9
Full unbal	s1	94.0	61.7	67.4
	s2	96.0	45.2	75.3
	s3	94.2	63.7	66.5
	s4	94.4	64.4	66.1
Full bal	s1	84.6	85.3	50.1
	s2	82.2	83.7	48.6
	s3	84.4	83.3	51.3
	s4	82.9	83.7	49.4
	s5	84.4	84.0	50.3

Table 12: Per-seed results for Qwen2.5 GRPO. Balanced training consistently keeps position-swap consistency high and the first-position rate near 50%.

J Statistical Uncertainty Checks

We add local uncertainty checks for the main diagnostic effects (Table 13). For single-run SFT comparisons, we use paired bootstrap resampling over the 449 matched test examples. For multi-seed GRPO comparisons, we bootstrap the available seed means; these intervals are descriptive because the number of seeds is small.

Comparison	Metric	Diff	95% CI
SFT unbal → SFT bal	Con	+87.1	[84.0, 90.0]
SFT unbal → SFT bal	First-pos	-48.9	[-50.6, -47.2]
Baseline → SFT bal	Avg	+12.5	[9.5, 15.6]
Baseline → SFT bal	Con	+5.6	[1.1, 10.0]
Qwen2.5 GRPO unbal → bal	Con	+25.3	[19.8, 34.4]
Qwen2.5 GRPO unbal → bal	Avg	+7.8	[5.3, 11.6]
Qwen3 GRPO unbal → bal	Con	+2.2	[1.1, 3.3]
Mistral GRPO unbal → bal	Con	+45.2	[40.4, 50.1]

Table 13: Bootstrap uncertainty checks. Diff is after − before in percentage points. Single-run SFT rows use paired-instance bootstrap; multi-seed GRPO rows use seed-mean bootstrap.

The intervals support the paper's qualitative interpretation without requiring stronger statistical language. The SFT shortcut collapse and recovery are much larger than instance-level uncertainty, including a 12.5-point gain in order-averaged accuracy over the baseline. For Qwen2.5 GRPO, balancing improves both consistency and order-averaged accuracy; the latter gain is 7.8 points with a descriptive seed-mean interval of [5.3, 11.6]. For Mistral GRPO, the consistency gain is large, whereas the Qwen3 gain is small but positive under the local bootstrap, matching the conclusion that shortcut severity is model- and configuration-dependent.

K Learning Rate Sensitivity

The complete learning rate data is shown in Table 14 and visualized in Figure 5.

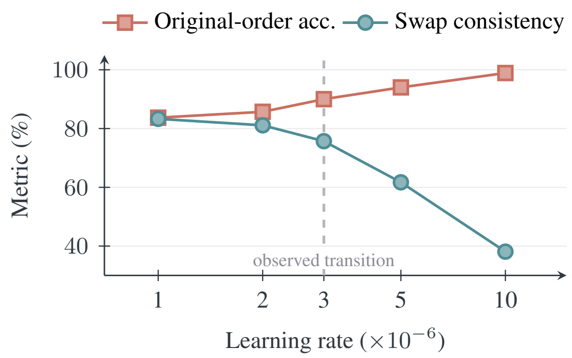

Figure 5: Learning-rate sensitivity in the single-seed Qwen2.5 full-reward sweep. Original-order accuracy rises as position-swap consistency falls; no uncertainty is shown because each point is one run. The dashed marker denotes an observed transition region in this grid, not an estimated universal threshold.

Learning Rate	Acc	Con	First-pos
Baseline	80.2	81.5	49.3
1e-6	83.7	83.3	51.0
2e-6	85.7	81.1	54.3
3e-6	90.0	75.7	58.2
5e-6 (standard)	94.0	61.7	67.4
1e-5	98.9	38.1	80.5

Table 14: Full learning rate sweep. First-position rate increases with training intensity as consistency falls.

L Training Algorithm Controls

The position shortcut is not specific to any training paradigm. SFT on unbalanced data achieves 100% accuracy with 0% consistency. DPO falls between SFT and GRPO in severity (94.2%/54.3%). GRPO retains partial consistency (58.7% multi-seed mean). The observed ordering SFT > DPO > GRPO describes shortcut severity in this configuration, but it does not isolate an optimizer-level mechanism; all three objectives remain vulnerable to the position-label confound.

When given balanced data, SFT achieves higher original-order accuracy (91.3% versus 84.6% for GRPO in the corresponding single run), showing that direct supervision can use deconfounded labels effectively in this setting. The same SFT procedure collapses under unbalanced labels, so this comparison supports sensitivity to data construction rather than a claim about relative data efficiency or optimizer mechanism.

M Cross-Model Learning Rate Sensitivity

Figure 6 shows a much flatter learning-rate curve for the tested Qwen3-8B accuracy-reward configuration. Since the Qwen2.5 curve uses the full reward, the comparison does not isolate model family from reward setup.

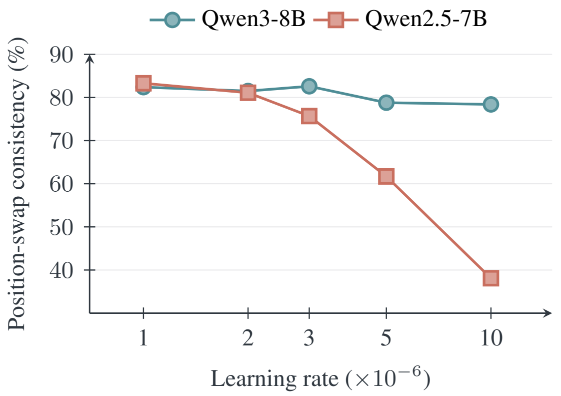

Figure 6: Position-swap consistency across two single-seed learning-rate sweeps. Qwen2.5-7B loses about 45 pp under the full reward, whereas Qwen3-8B changes by about 4 pp under the accuracy reward. Each point is one run, and the differing rewards preclude a model-only causal comparison.

The contrast shows that shortcut sensitivity is not constant across tested configurations. Qwen3-8B maintains higher consistency under moderate learning rates in its accuracy-reward sweep, but its curve should not be interpreted as evidence that scaling caused the difference. The practical conclusion is narrower: every model and training recipe still requires a position-swap audit.

N Multi-Objective Reward Analysis

A key question is whether reward engineering can substitute for data deconfounding. We test three reward configurations on both models (all on unbalanced data). For Qwen2.5-7B, rows report seed means where available; Qwen3-8B rows are single-seed reward controls.

Reward	Qwen2.5-7B		Qwen3-8B	
	Acc	Con	Acc	Con
Accuracy only	94.7	59.8	88.4	78.8
+ Decisive	93.9	66.3	88.2	82.0
+ Confidence proxy	95.4	56.1	89.1	78.0
Full composite	94.7	58.7	87.8	80.8

Table 15: Reward ablation across models. Qwen2.5 rows use multi-seed means where available, while Qwen3 rows are single-seed controls. No proxy reward exceeds balanced data (Qwen2.5 GRPO balanced: 84.0% con, Qwen3 GRPO balanced: 82.2% con). The fixed-confidence proxy lowers consistency on both models.

The tested proxy rewards do not replace data deconfounding (Table 15). The decisiveness proxy penalizes tie hedging but cannot distinguish content-based decisiveness from following the position shortcut. The fixed-confidence proxy lowers consistency on both models because, with verdict-only outputs, it is almost a deterministic transformation of the predicted label and correctness rather than an independent calibration signal. Overall, these components help only partially and do not match the direct intervention of removing the position-label correlation from the training data.

O Training Dynamics Details

Table 16 gives the paired checkpoint-level data for the training dynamics experiment. At these shared checkpoints, original-order accuracy and the first-position rate increase as training proceeds, while consistency is preserved only at the earliest checkpoint and then drops sharply. The composite reward follows the same shortcut trajectory rather than materially slowing the collapse.

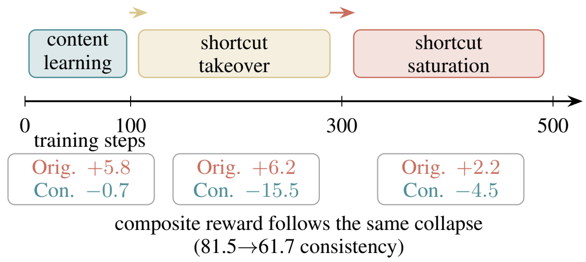

Figure 7: Single-seed checkpoint dynamics for accuracy-only GRPO. Early training raises original-order accuracy with little consistency loss; the middle checkpoints show shortcut takeover, and the final phase approaches a position-dependent policy. Phase boundaries summarize the observed checkpoints rather than estimated change points.

Steps	Acc-only			Full		
	Acc	Con	First	Acc	Con	First
0 (base)	80.2	81.5	49.3	80.2	81.5	49.3
100	86.0	80.8	53.8	84.6	81.7	53.5
200	89.1	76.8	57.1	90.9	74.6	59.8
300	92.2	65.3	63.7	93.5	65.9	64.9
500	94.4	60.8	67.4	94.0	61.7	67.4

Table 16: Training dynamics at paired checkpoints. Original-order accuracy and the first-position rate increase together, while position-swap consistency drops after the earliest checkpoint. The composite reward reaches nearly the same endpoint as the accuracy-only reward.

P Per-Domain Analysis

Each domain's vulnerability to the shortcut varies substantially (Table 17). We report the primary Qwen2.5 GRPO full setting by domain, using seed means for the unbalanced and balanced rows. The shortcut appears in every domain, but reasoning shows the cleanest collapse: unbalanced GRPO reaches 98.6% original-order accuracy while consistency falls to 19.6% and the first-position rate rises to 90.2%. Balanced training moves this rate back toward neutral and recovers consistency in all domains. Because the smallest slices contain only 43–69 examples, these results are diagnostic rather than independent domain conclusions.

Setting	Metric	Chat	Reas.	Safe	Code	Adv.
Baseline	A	96.6	78.3	80.9	85.2	32.6
	C	93.1	63.8	90.4	78.5	72.1
	F	51.1	60.1	48.3	45.2	44.2
Unbal.	A	99.4	98.6	94.1	96.3	75.0
	C	64.9	**19.6**	73.7	64.4	51.2
	F	67.1	**90.2**	62.5	63.0	73.3
Bal.	A	94.3	85.5	82.6	91.3	38.6
	C	91.5	70.7	88.7	84.1	77.2
	F	50.6	58.6	48.8	47.9	44.4

Table 17: Per-domain Qwen2.5 GRPO full diagnostics. A is original-order accuracy, C is position-swap consistency, and F is the two-order first-position rate. Unbalanced and balanced rows report seed means. Domain sizes are Chat 87, Reasoning 69, Safety 115, Code 135, and Adversarial 43.

Q Stylized Slot-Bias Derivation

The functional argument in the main text can be made explicit with a deliberately simple discriminative model. Let $y \in \{+1, -1\}$ denote whether the first or second response is preferred, and write the judge logit as

$$f_\theta(x) = d_\theta(x) + b,$$

where $d_\theta(x)$ summarizes content-comparison evidence and b is a global first-position bias. Under logistic loss,

$$\mathcal{L} = \mathbb{E}_{(x,y)} \left[\log(1 + \exp(-y(d_\theta(x) + b)))\right],$$

the derivative with respect to the position term is

$$\frac{\partial \mathcal{L}}{\partial b} = -\mathbb{E}_{(x,y)} \left[y\, \sigma(-y(d_\theta(x) + b))\right].$$

Near initialization, with $d_\theta(x) \approx 0$ and $b \approx 0$, $\sigma(0) = 1/2$. If $\alpha = \Pr(y = +1)$, then

$$\frac{\partial \mathcal{L}}{\partial b} \approx -\frac{1}{2}\mathbb{E}[y] = -\frac{1}{2}(2\alpha - 1).$$

When every preferred answer is in the first slot ($\alpha = 1$), gradient descent therefore increases b. When positions are balanced ($\alpha = 0.5$), this first-order global shortcut direction vanishes. This calculation is an explanatory local model, not a GRPO derivation or evidence that the trained transformer has one scalar bias.

R Confound-Ratio and Data Duplication Control

We verify that the balanced training fix does not depend on data duplication. Our standard balanced construction creates a mirror of each instance with swapped positions, doubling the dataset from 2,089 to 4,178 training pairs. To control for this size difference, we also train on non-duplicated balanced data (2,089 instances, each randomly assigned to either its original or swapped position with equal probability). The non-duplicated run yields near-identical original-order accuracy and position-swap consistency to the mirrored balanced seed mean (83.1%/82.6% versus 83.7%/84.0%), with the first-position rate still near neutral (50.2%). Thus the improvement is attributable to deconfounding rather than data duplication.

We also train at intermediate confound ratios. Figure 8 reports consistency loss relative to the balanced reference as the share of position-A preferred examples increases from 50% to 100%. The 50% reference and 100% endpoint use multi-seed means, whereas the intermediate ratios are single-seed diagnostics. Those intermediate values are non-monotonic, so we interpret the sweep as a boundary diagnostic rather than evidence of a smooth dose-response relation. The clearest result is the sharp deterioration at the fully unbalanced endpoint.

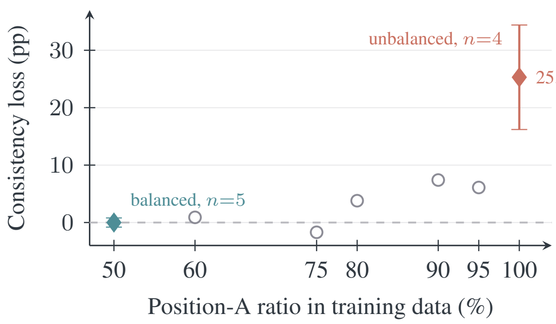

Figure 8: Confound-ratio diagnostic measured as consistency loss from the balanced reference. Diamonds are multi-seed endpoint means with seed standard deviation; hollow circles are single-seed intermediate diagnostics and are intentionally not connected. The intermediate values are non-monotonic, while the fully unbalanced endpoint shows the sharp deterioration.

S Sharp-Transition Analysis

The single-seed Qwen2.5 learning-rate sweep shows a sharp transition within the tested grid rather than uniform degradation (Figure 5). At low rates, consistency remains close to the baseline; at larger rates, original-order accuracy rises while consistency falls. The sparse grid does not identify a critical threshold, but shows that conclusions at one learning rate need not transfer.

The Qwen3-8B accuracy-reward sweep is flatter, maintaining high-70s to low-80s consistency and reaching 78.4% at the largest rate (Table 10). Because the reward setups differ, this is a robustness check rather than a direct attribution to model capability. Practitioners should therefore audit each deployed model–reward–learning-rate combination.